

TDT4186 - sample exam

1 Multiple-Choice Questions

1. Which of the following about process is **not correct**?
 - A. Process is a running program.
 - B. Every process has its own page table.
 - C. Every process maintains a structure to store all opened files.
 - D. Every process maintains a structure to store all locks it is using.
2. What is the output of the following code snippet? (Assume that process creation succeeds, so no error check.)

```
ret = fork();
printf("A-");
if (ret > 0){
    wait();
    printf("B-");
}
else {
    printf("C-");
}
```

- A. A-B-C-
 - B. A-C-B-
 - C. A-A-B-C-
 - D. A-A-C-B-
3. The following task set is scheduled using STCF, what is the turnaround time of P1?

	Arrival Time	Execution Time
P1	0	30
P2	10	15
P3	20	10

- A. 30
- B. 45
- C. 50
- D. 55

4. Which of the following scheduling algorithm maintains a circular queue?
- A. FIFO
 - B. SJF
 - C. STCF
 - D. RR
5. Which of the following one is **correct** for CFS scheduling algorithm in Linux?
- A. The complexity of selecting the next scheduled task for CFS is $O(\log(n))$
 - B. CFS always schedules the task with the minimum *vruntime*
 - C. The design objective of CFS is to improve average response time
 - D. CFS deploys a B+ tree structure to store *vruntime* of all tasks
6. Which of the following one is **not correct**?
- A. Segmentation may cause external fragmentation.
 - B. Each segment has two registers, one to store the starting address of the segment and another to store the ending address of the segment.
 - C. Paging may cause internal fragmentation.
 - D. Using TLB can reduce the number of memory accesses.
7. If a process has m different pages, the minimum number of page fault is
- A. m
 - B. 0
 - C. 1
 - D. $m - 1$
8. When launching a process, an address space is created for this process. Which of the following one is **correct** about address space?
- A. For all OSes, address space only consists of three parts, i.e., code, heap, and stack
 - B. Heap and stack within address space can grow dynamically at runtime.
 - C. Heap is used to store local variables.
 - D. Stack is used to store dynamically-allocated data structures.
9. Assuming that a process has 4 different pages, i.e., A, B, C, and D. Physical memory has 3 frames and the page access sequence is A-B-A-C-D. When accessing page D, which page will be replaced using LRU replacement policy? And how many page faults will the given page sequence have?
- A. Replace page A; 3 page faults
 - B. Replace page B; 4 page faults

- C. Replace page C; 3 page faults
 - D. Replace page A; 4 page faults
10. Which of the following is **not** the requirement of solutions for critical-section problems?
- A. Mutual Exclusion
 - B. Non-preemption
 - C. Progress
 - D. Bounded Waiting
11. Which of the following is **not** a condition for deadlock?
- A. Mutual Exclusion
 - B. Hold and Wait
 - C. Full Preemption
 - D. Circular Wait
12. Which of the following statement is **true** regarding memory-mapped I/O?
- A. The CPU accesses the device memory much like it accesses main memory.
 - B. The CPU uses separate architecture-specific instructions to access memory in the device.
 - C. Memory-mapped I/O can be used only with a polling-based device driver.
 - D. Memory-mapped I/O can be used only with an interrupt-driven device driver.
13. Which of the following concept is **not** a part of hard disk drives?
- A. Cylinder
 - B. Cell
 - C. Sector
 - D. track
14. Processes 1, 2, and 3 read a file `test.txt` using `open()+read()`, then which of the following statement is **correct**?
- A. Processes cannot have same file descriptor number for `test.txt`.
 - B. If one process changes the offset of its file descriptor, all other file descriptors will update offsets accordingly.
 - C. All processes will have different inodes for file `test.txt`.
 - D. None of the above.
15. Which of the following is **correct** for file systems?
- A. Superblock includes all information of free inodes blocks and data blocks.

- B. Directories are a specific-design data structure different from files.
 - C. If an inode block is released, it can be reused for another file.
 - D. Bitmap can only be used for data blocks.
16. When reading a file `/root/bar.txt`, we need to access several parts of the on-disk structures to complete this operation
1. inode block of `root`
 2. inode block of `bar`
 3. data block of `root`
 4. data block of `bar`
- what is the correct access order?
- A. 1, 2, 3, 4
 - B. 1, 3, 4, 2
 - C. 1, 3, 2, 4
 - D. 2, 1, 4, 3

2 Multi-Parts Questions

1. Given the following snippet,

```
int count = 0;
int i;
for (i=0; i<2; i++) {
    fork();
    count++;
}
printf("%d\n", count);
```

- (a) (1 point) How many processes will this snippet create at the end?
 - (b) (1 point) What will the above snippet print out?
2. Paging deploys a per-process page table to store the information of translating page numbers to frame numbers for each process.
- (a) (1 point) Where are page tables stored? How many memory accesses does a virtual address need to point to the corresponding physical memory in page-based memory management without TLB?
 - (b) (1 point) What is the purpose of TLB? If a page entry is not found in TLB, i.e., page fault, how does OS handle a page fault?

- (c) (2 points) What is LRU in page replacement? And how does the real system implement LRU?
3. (a) (1 point) What is the hybrid lock? How does it work?
- (b) (2 points) Can we implement a lock using semaphore? Assume `critical_sect()` is the critical section we are going to have mutual exclusion. Explain how to initialize semaphore and where should the two operations be added for the critical section?
4. Two threads access variable `x`, and their assembly code is as follows.

Thread 1	Thread 2
<code>mov R1, x // x -> R1;</code>	<code>mov R2, x // x -> R1;</code>
<code>inc R1 // R1 + 1 -> R1</code>	<code>inc R2 // R1 + 1 -> R1</code>
<code>mov x, R1 // R1 -> x</code>	<code>mov x, R2 // R2 -> x</code>

where `R1` and `R2` are two registers. Assume that `x = 2` and both threads only manipulate `x` once.

- (a) (1 point) How many different values can we have for `x` ?
- (b) (2 points) What are the values? How does the execution end up getting them?
5. Given an OS with a paging-based memory management, the system allows for **1GB** virtual address spaces for processes. The page size is **4KB**, and the system has **4 GB** of physical memory.
- (a) (1 point) How many physical frames does the system have?
- (b) (1 point) How many bits does the virtual address have for virtual page number?
- (c) (1 point) If the size of page table entry (PTE) is 4 bytes and it has two-level page table, how many bits does it use to point to the first-level page directory?
- (d) (1 point) Consider a process references to 11 different virtual addresses and TLB can store 10 entries. The process repeatedly accesses the 11 virtual addresses in a fixed order, what will be the hit rate of TLB when using FIFO as the page replacement policy? What will be the hit rate of LRU?
6. Three threads access two shared semaphores as follows:

Thread 1	Thread 2	Thread 3
<code>while (1){</code>	<code>while (1){</code>	<code>while (1){</code>
<code>sem_wait(U);</code>	<code>sem_wait(V);</code>	<code>sem_wait(V);</code>
<code>printf("A");</code>	<code>printf("B");</code>	<code>printf("C");</code>
<code>sem_post(V);</code>	<code>sem_post(V);</code>	
<code>}</code>	<code>}</code>	<code>}</code>

where `U` is initialized to 3 and `V` is initialized to 0.

- (a) (1 point) If execution gets stuck, how many "A" will be printed out? Explain the reason.
- (b) (1 point) What is the minimal number of printed "B"? Explain the reason.
- (c) (1 point) Is "ACBABCABC" a possible output sequence? Explain the reason.